# Free-body diagrams, part 1

**LEARNING TARGET · S5** I can draw a complete FBD — every force, with arrows sized to strength.

## A · KEY VOCABULARY

| | |
|---|---|
| **Free-body diagram** | Only the forces acting *on* one chosen object. |
| **The dot / box** | Replace the object with a simple shape; draw arrows from it. |
| **Source** | Every arrow needs one: rope, floor, table, Earth. |
| **No "motion force"** | Motion is not an interaction. Leave velocity off the diagram. |

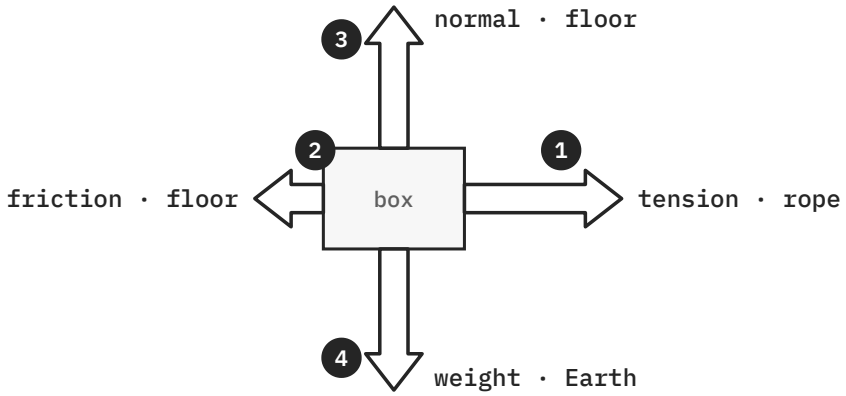
## B · SIX FBD RULES

1 just the object (a dot)
2 one arrow per force
3 arrow points the force's way
4 length ∝ strength
5 label each: $F_g$, $F_N$, $F_f$…
6 only forces ON the object

**routine** choose · list · draw

## C · COLOR IT · A BOX PULLED RIGHT

*Rough level floor, rope pulls right.*




- ☐ **1 · ORANGE** Tension (T)
- ☐ **2 · ORANGE** Friction (f)
- ☐ **3 · ORANGE** Normal force (N)
- ☐ **4 · ORANGE** Weight (W)

## D · WORKED EXAMPLE

A 20 N lamp hangs at rest from one vertical rope. Draw and label its forces.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| weight = 20 N<br>at rest<br>one rope | sources: Earth, rope | weight 20 N down<br>tension up, balances it | **Tension 20 N up**<br>equal-length arrows |

## E · THE 2026-XJ THREAD

DAY 15 OF 22 · NEW TOOL: FREE-BODY DIAGRAMS

A free-body diagram puts XJ on a dot and shows every force acting on it. For an object in space the list is short, which is what makes it predictable.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else